

Introduction to Machine Learning

Machine Learning (Bilingual)

齐飞 (Fei **Qi**) <fred.qi@ieee.org>

李阳 (Yang **Li**)

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School of Artificial Intelligence
Xidian University





Basic Concepts of Machine Learning

Types of Machine Learning Algorithms

Machine Learning Issues

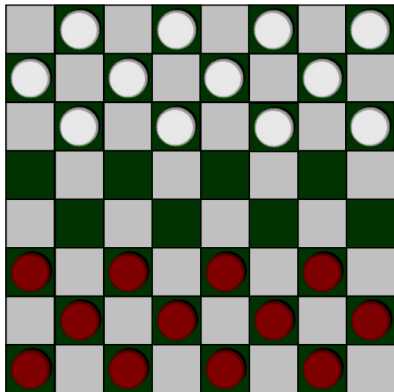
Summary

Basic Concepts of Machine Learning

Definition of Machine Learning



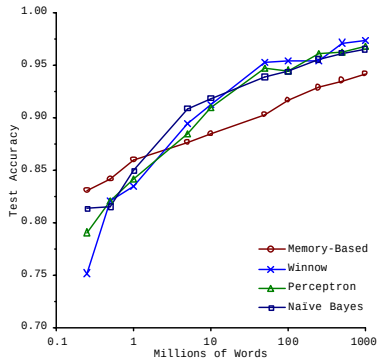
Machine Learning (Samuel 1959) Field of study that gives computers the ability to learn without being explicitly programmed.



Well-posed Learning Problem (Mitchell 1998) A computer program is said to **learn from experience E** with respect to some **task T** and some **performance measure P** , if its performance on T , as measured by P , improves with experience E .

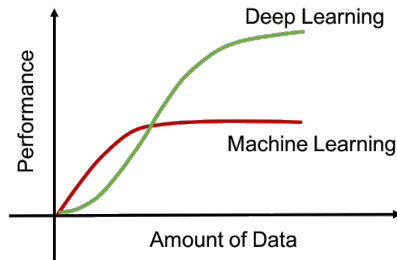
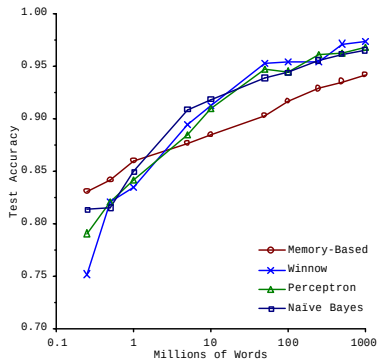
Suppose an email program can record whether you mark an email as spam and learn from this data to filter spam more effectively. In this problem, what is the **task T**?

- Classifying emails as spam or not spam
- Recording your tagging of emails as spam or not spam
- The number (or proportion) of emails correctly classified as spam or not spam
- None of the above: This is not a machine learning problem



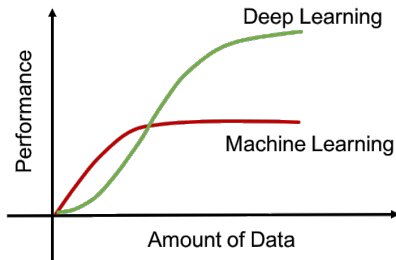
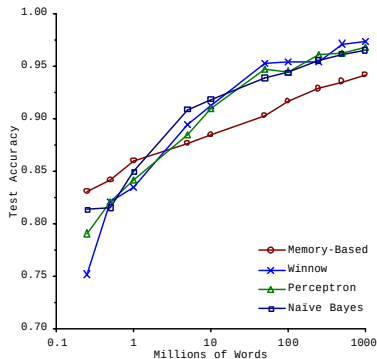
(Banko and Brill 2001)

Large-scale Machine Learning



(Banko and Brill 2001)

Large-scale Machine Learning



(Banko and Brill 2001)

“It’s not who has the best algorithm that wins.
It’s who has the most data.”

Types of Machine Learning Algorithms

Types of Machine Learning Algorithms



- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Recommender System



Supervised Learning involves working with known
“correct answers”

Regression predicts a continuous variable as output

Classification predicts a discrete variable as output (such as
0 and 1)

Supervised Learning involves working with known “correct answers”

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House Price Prediction

Supervised Learning involves working with known
“correct answers”

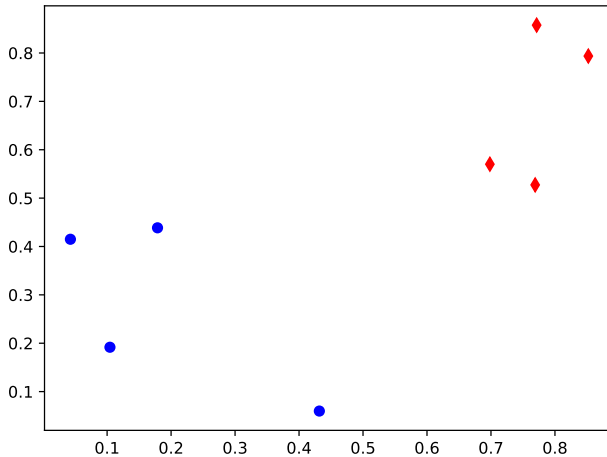
Regression predicts a continuous variable as output

Classification predicts a discrete variable as output (such as
0 and 1)

House Price Prediction

Tumor Diagnosis/Cancer Prediction

Classification Data Examples in Supervised Learning



Conceptual Clarification of Supervised Learning



P1: With several products each having numerous items, the goal is to predict the sales volume of these products over the next three months.

P2: Using software to check customer accounts, with each account determining whether it has been hacked or misappropriated.

Which of the above two problems are regression or classification problems:

- Both are classification problems.
- P1 is a classification problem, P2 is a regression problem.
- P1 is a regression problem, P2 is a classification problem.
- Both are regression problems.

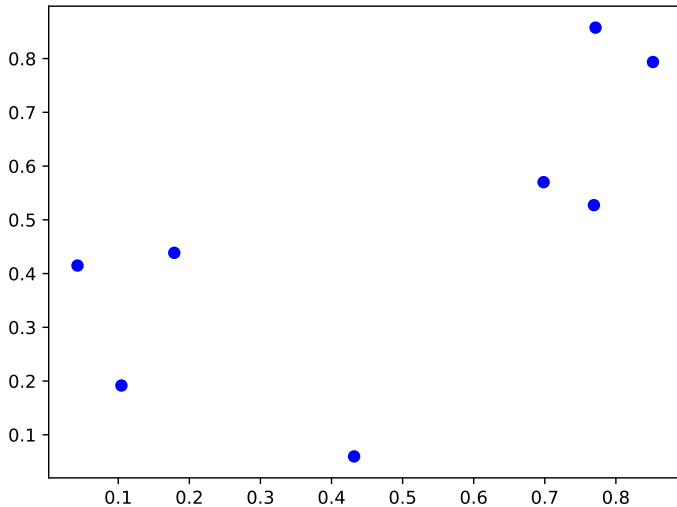


Clustering

Probabilistic Density Estimation

Dimensionality Reduction/Visualization

Samples of Unsupervised Learning

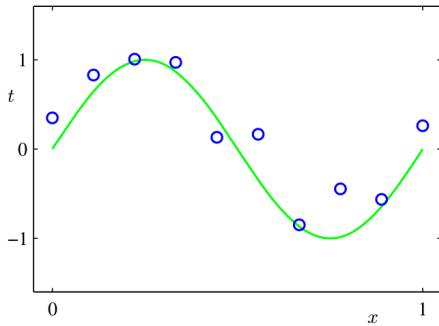


Among the following examples, which ones would you solve using **unsupervised learning** techniques?

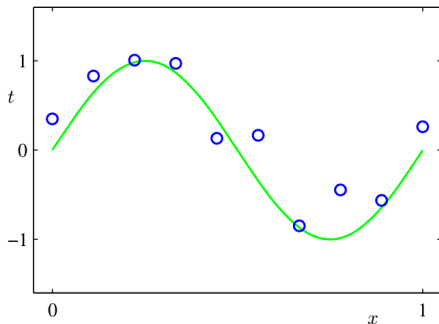
- Given emails and labels indicating whether they are spam, learn a spam filter.
- Given a set of news acquired from the web, classify them based on whether they are about the same event.
- Based on a database of customer data, automatically discover market segmentation and categorize customers into relevant market segments.
- With a database diagnosing whether a patient has diabetes, learn to diagnose new patients for diabetes.

Machine Learning Issues

Curve Fitting

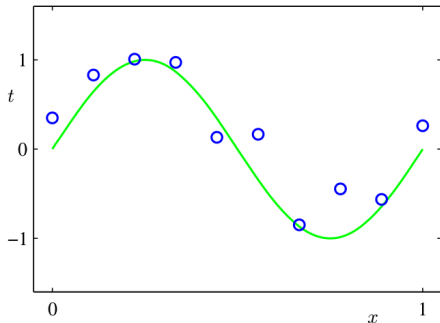


Curve Fitting



- The training set contains N observations about $\mathbf{x}$,
 $\mathbf{x} = (x_1, x_2, \dots, x_N)^T$.
- The corresponding target function values $\mathbf{t}$ are
 $\mathbf{t} = (t_1, t_2, \dots, t_N)^T$.

Curve Fitting

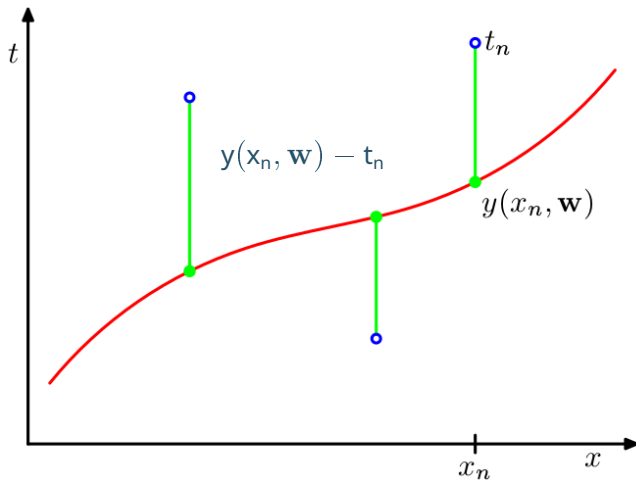


- The training set contains N observations about x , $\mathbf{x} = (x_1, x_2, \dots, x_N)^T$.
- The corresponding target function values t are $\mathbf{t} = (t_1, t_2, \dots, t_N)^T$.

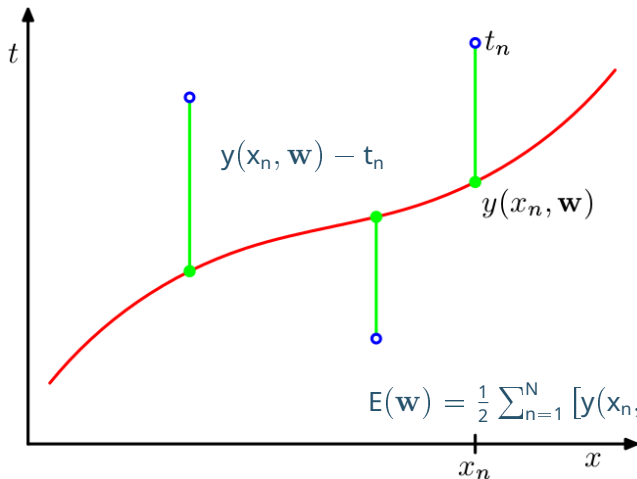
A polynomial function of the following form can be used for fitting

$$y(\mathbf{x}, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

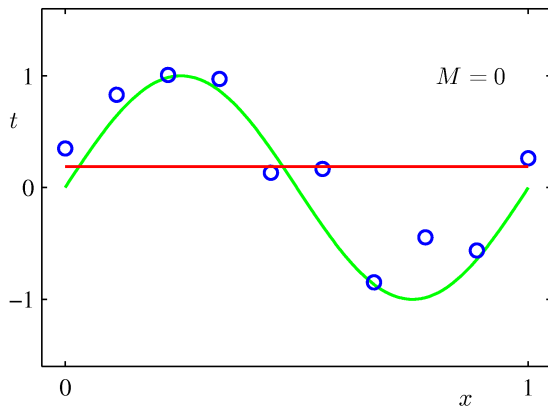
Error in Curve Fitting



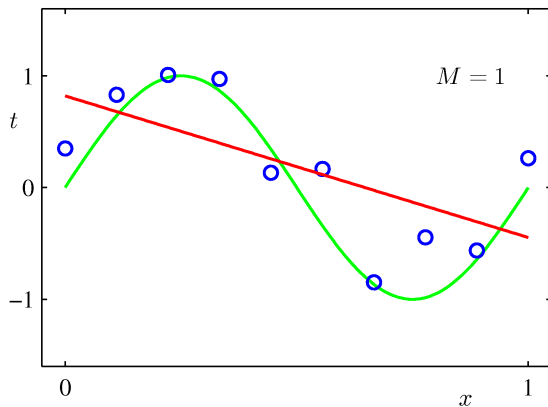
Error in Curve Fitting



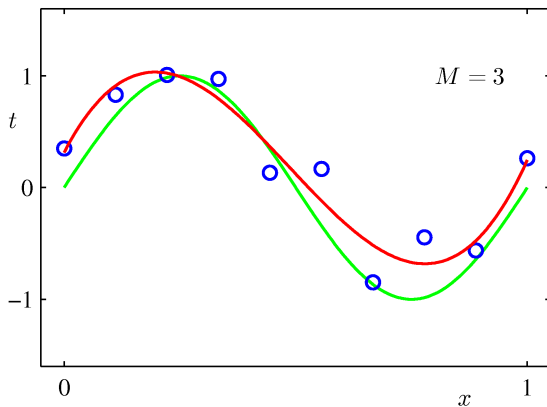
Model Order ($M = 0$)



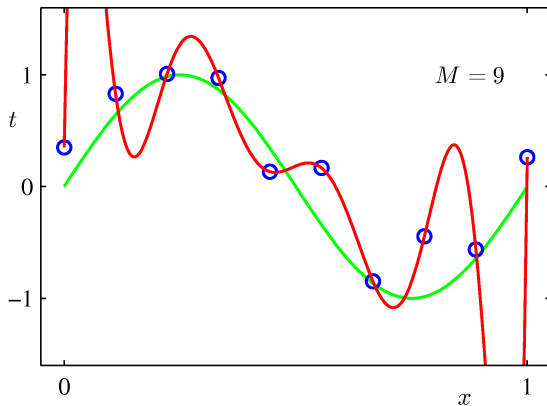
Model Order ($M = 1$)



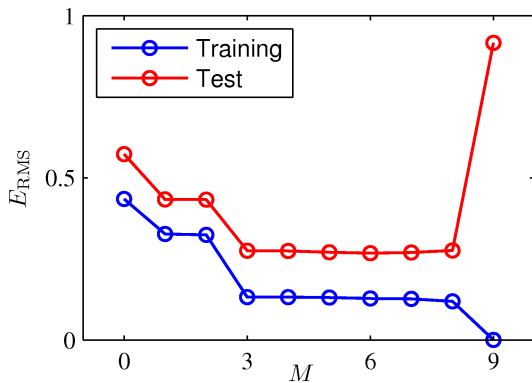
Model Order ($M = 3$)



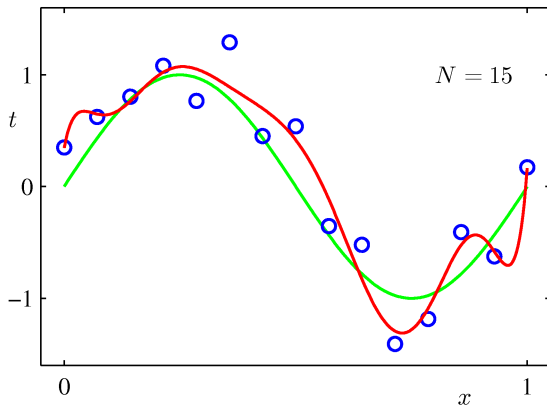
Model Order ($M = 9$)



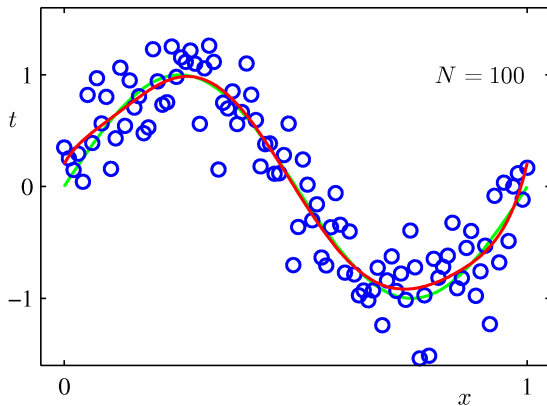
Overfitting



Size of Training Set ($N = 15$)



Size of Training Set ($N = 100$)



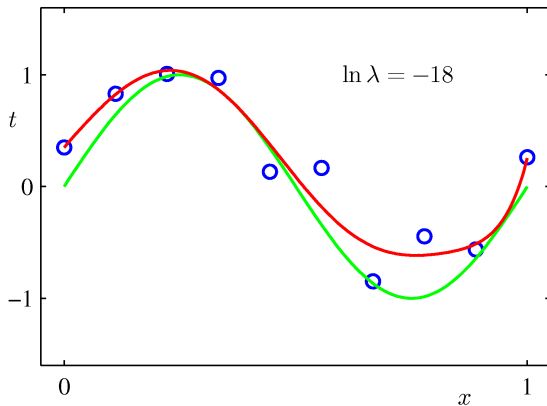
Polynomial Coefficients



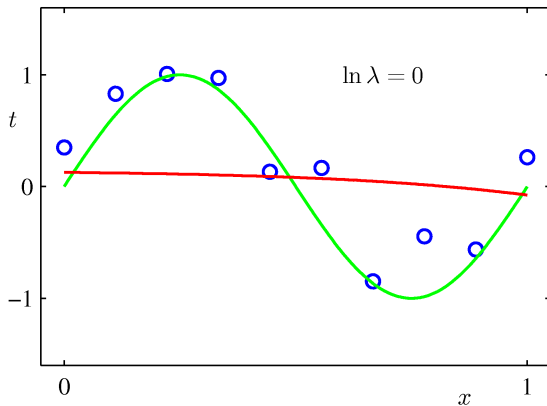
	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N [y(x_n, \mathbf{w}) - t_n]^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

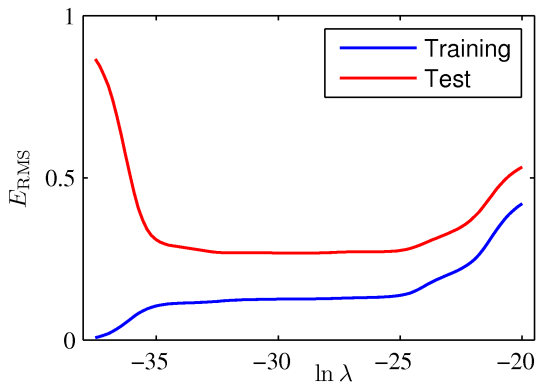
Regularization ($\ln \lambda = -18$)



Regularization ($\ln \lambda = 0$)



Regularization



	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
w_0^*	0.35	0.35	0.13
w_1^*	232.37	4.74	-0.05
w_2^*	-5321.83	-0.77	-0.06
w_3^*	48568.31	-31.97	-0.05
w_4^*	-231639.30	-3.89	-0.03
w_5^*	640042.26	55.28	-0.02
w_6^*	-1061800.52	41.32	-0.01
w_7^*	1042400.18	-45.95	-0.00
w_8^*	-557682.99	-91.53	0.00
w_9^*	125201.43	72.68	0.01

Summary

Definition of Machine Learning

Supervised and Unsupervised Learning

- Regression/Classification
- Clustering/Dimensionality Reduction/Visualization

Model Order and Regularization

- Overfitting
- Regularization

-  Banko, Michele and Eric Brill (July 6, 2001). **“Scaling to Very Very Large Corpora for Natural Language Disambiguation”**. In: *Proceedings of the 39th Annual Meeting on Association for Computational Linguistics*. ACL '01. USA: ACL, pp. 26–33. DOI: 10.3115/1073012.1073017.
-  Mitchell, Tom (1998). ***Machine Learning***. McGraw Hill.
-  Samuel, A. L. (July 1959). **“Some Studies in Machine Learning Using the Game of Checkers”**. In: *IBM Journal of Research and Development* 3.3, pp. 210–229. ISSN: 0018-8646. DOI: 10.1147/rd.33.0210.